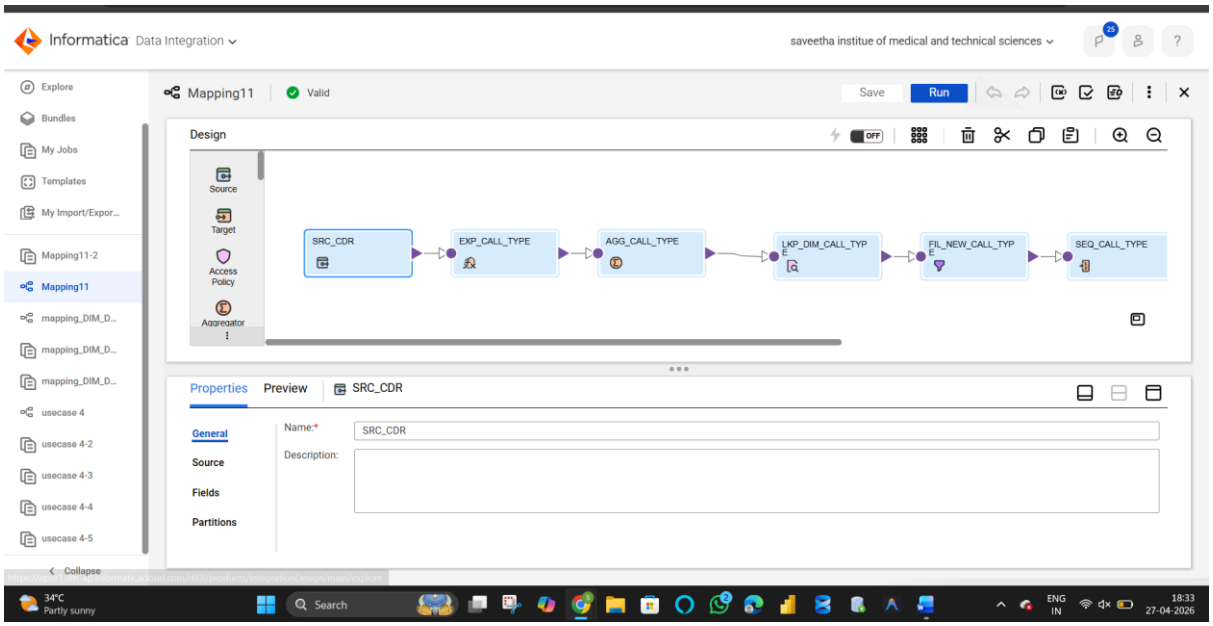
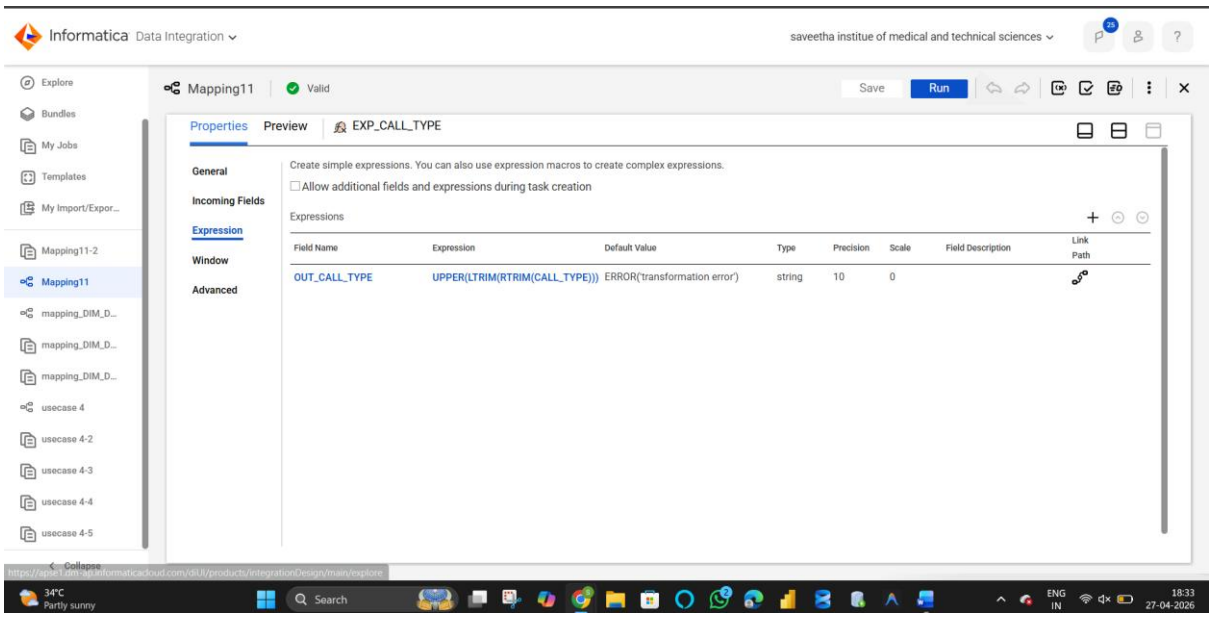
HCL -HACKTHON

M.SAINIKHITHA

192225110

Documentation – DIM_CALL_TYPE, STAGING & KPI (Use Case 4)





This taskflow is designed to process telecom call data through a structured ETL pipeline. The data is first loaded into the **staging layer**, then transformed and stored in the **DIM_CALL_TYPE dimension**, and finally used to compute KPIs for **Use Case 4**.

The main goal is to ensure data quality, proper classification of call types, and meaningful business insights.

2. Staging Layer – Description

The **staging layer** acts as an intermediate storage area where raw data is initially loaded.

Purpose:

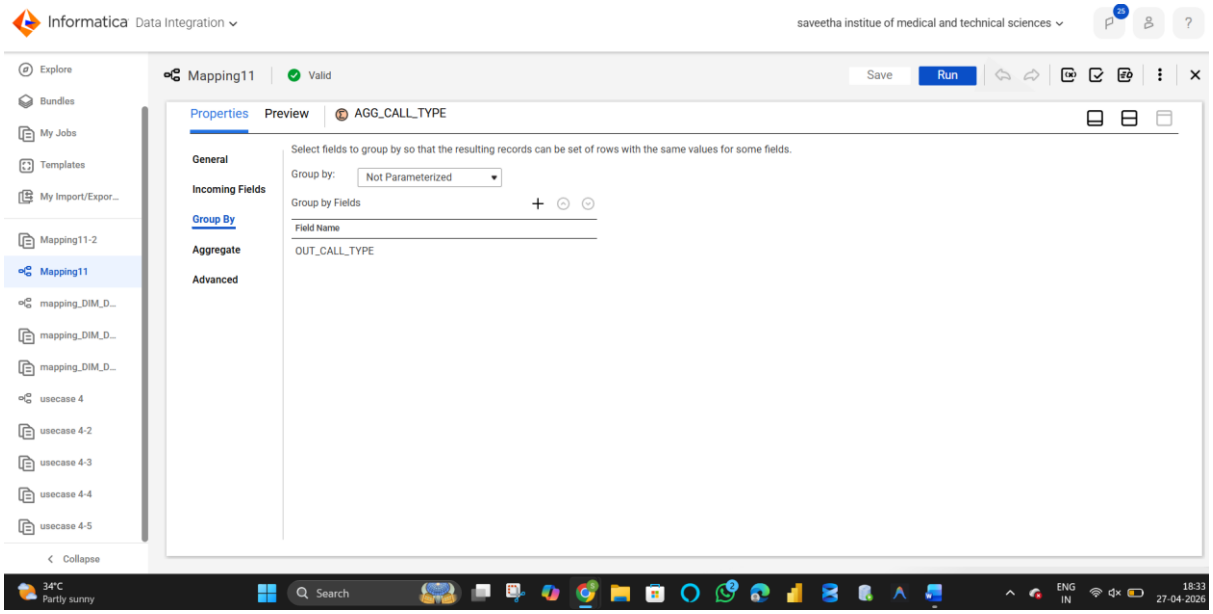
- To store raw telecom data before transformation
- To perform initial data validation and cleansing
- To isolate source system from transformation logic

Data Contains:

- CALL_ID
- CUSTOMER_KEY
- CALL_TYPE
- DURATION
- REVENUE
- DATE

Processing Steps:

- Data is extracted from source files or tables
- Basic validation is performed (null checks, format checks)
- Data is prepared for transformation into dimension and fact tables



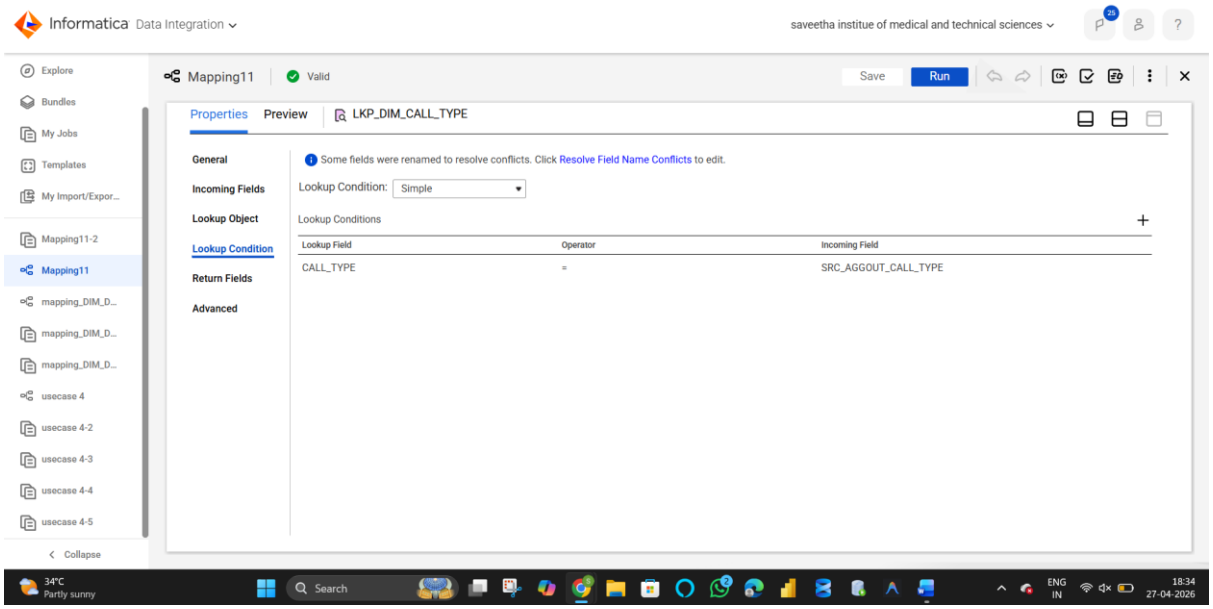
4. DIM_CALL_TYPE – ETL Logic

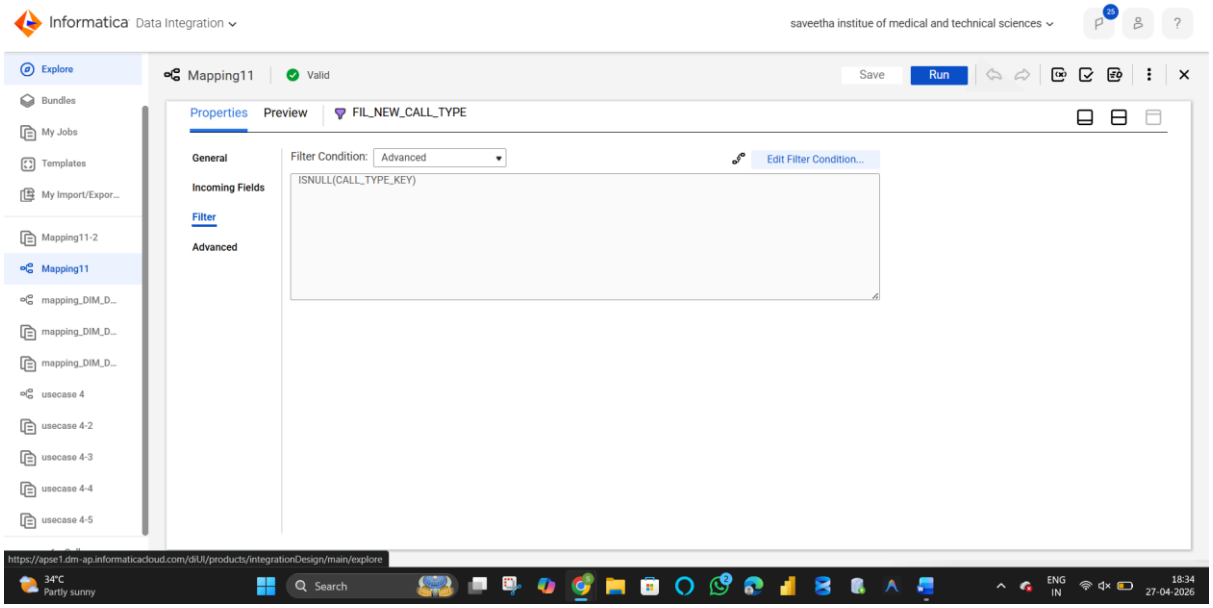
- Source data is read from staging tables
- **Expression transformation** is used to clean and standardize call type values
- **Lookup transformation** is used to check if the call type already exists
- If not found → new record is inserted
- If found → existing record is reused
- This ensures consistency using **dimension keys**

5. Transformations Used

- **Source Qualifier** – Reads data from staging
- **Expression Transformation** – Cleans and formats data
- **Lookup Transformation** – Checks existing call types

- **Router Transformation** – Filters valid and invalid records
- **Target Load** – Loads data into DIM_CALL_TYPE

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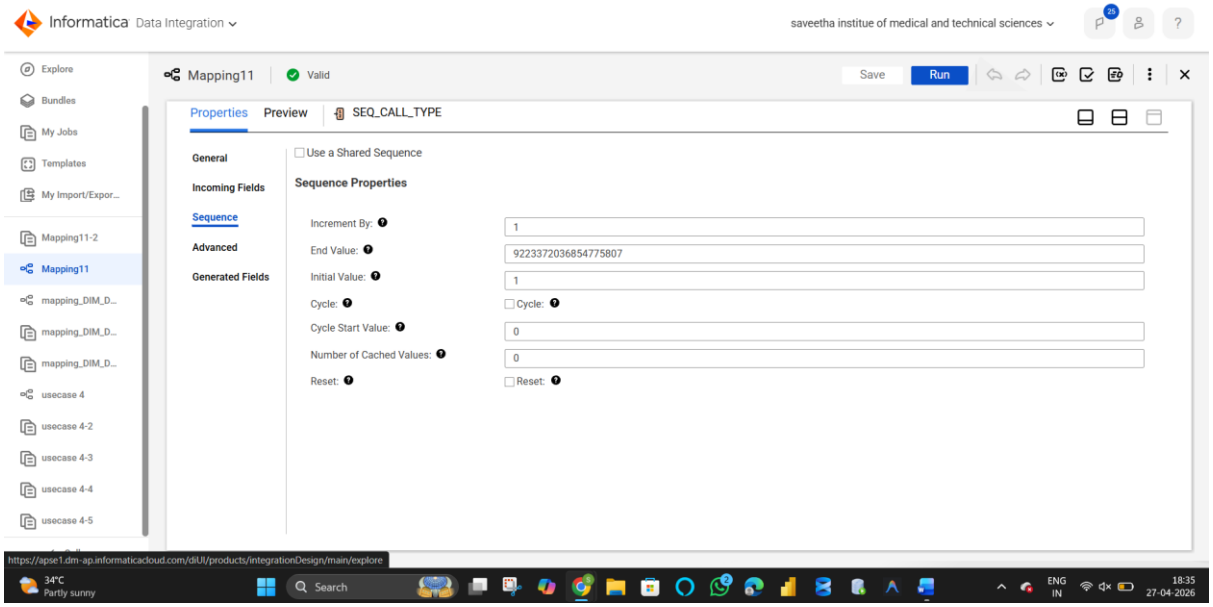
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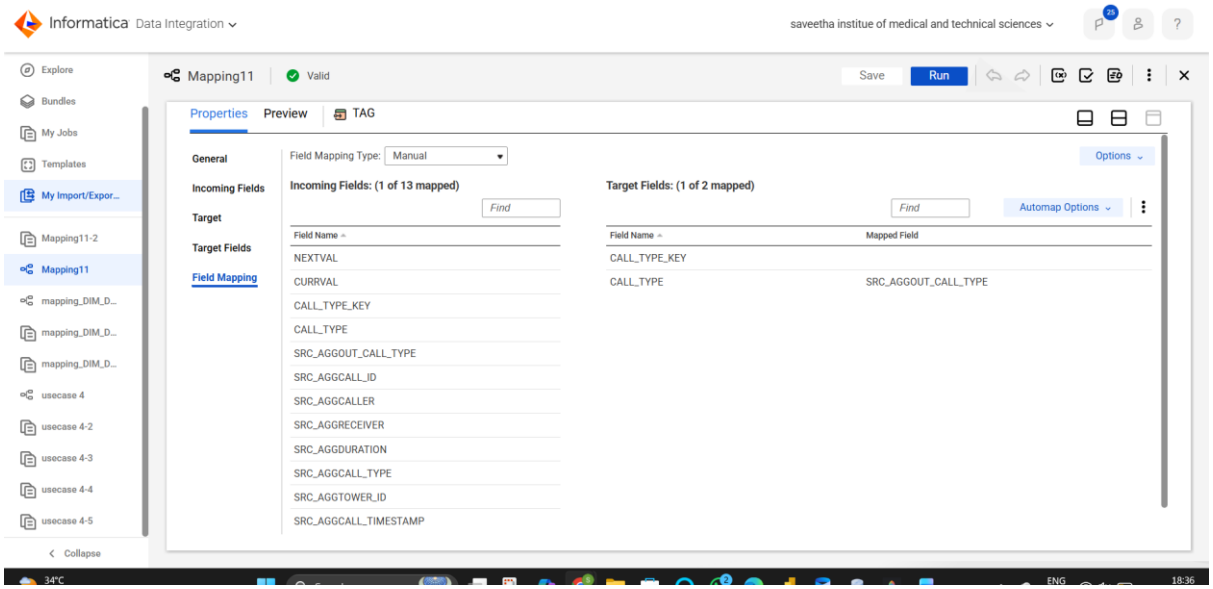
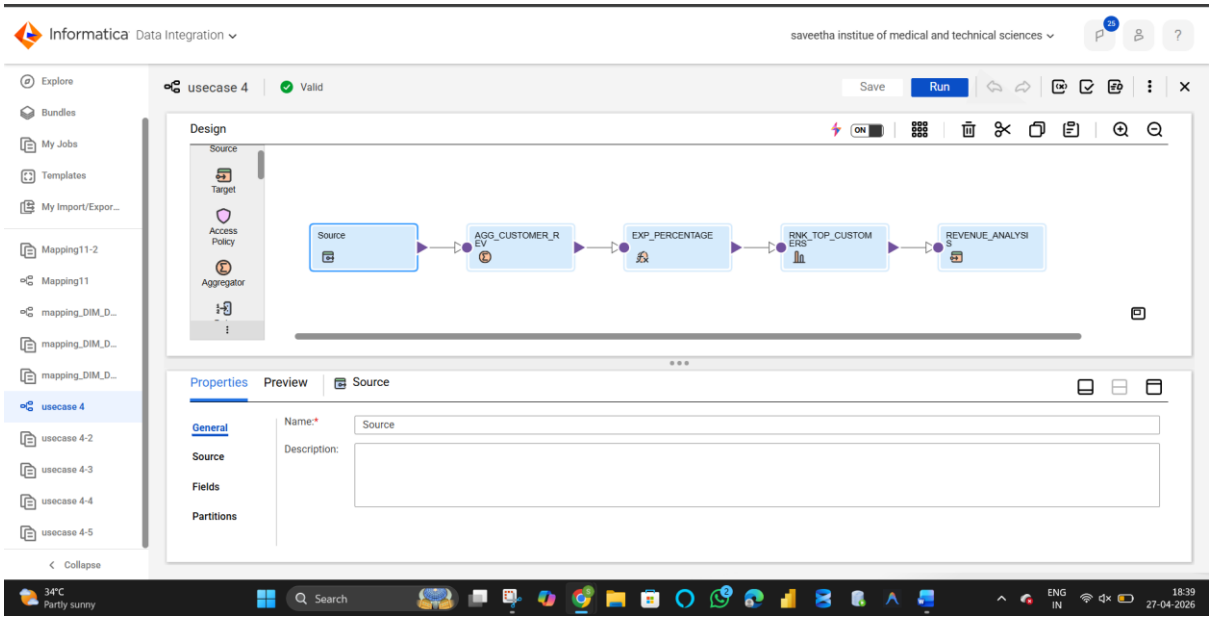


6. Use Case 4 – KPI Description

Use Case 4 focuses on analyzing telecom performance based on call data.

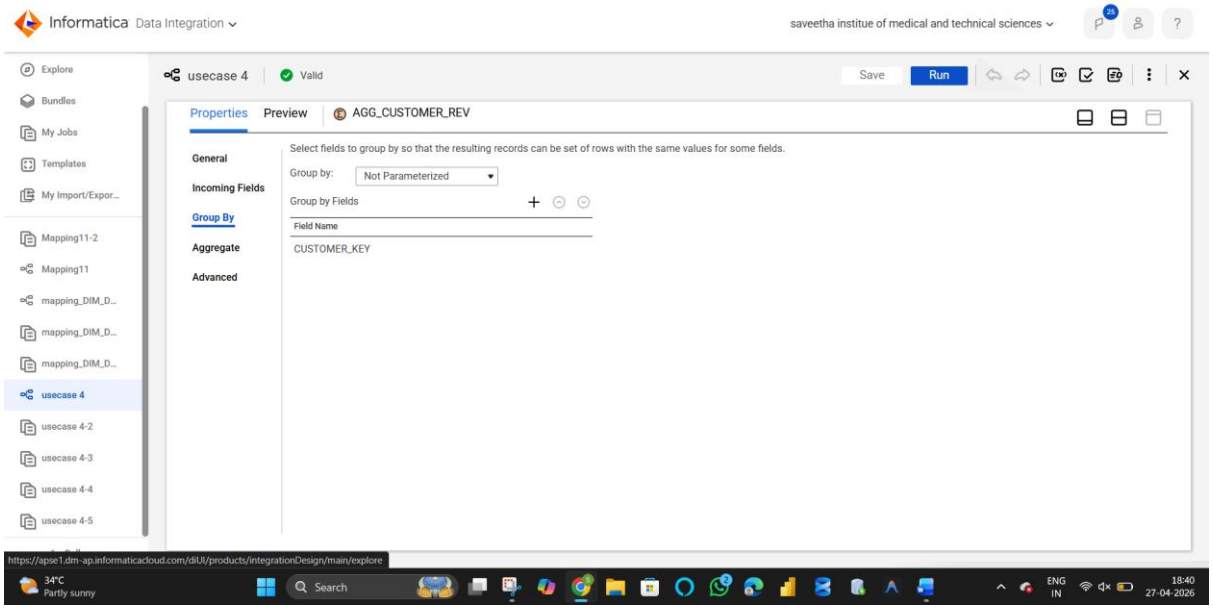
Objectives:

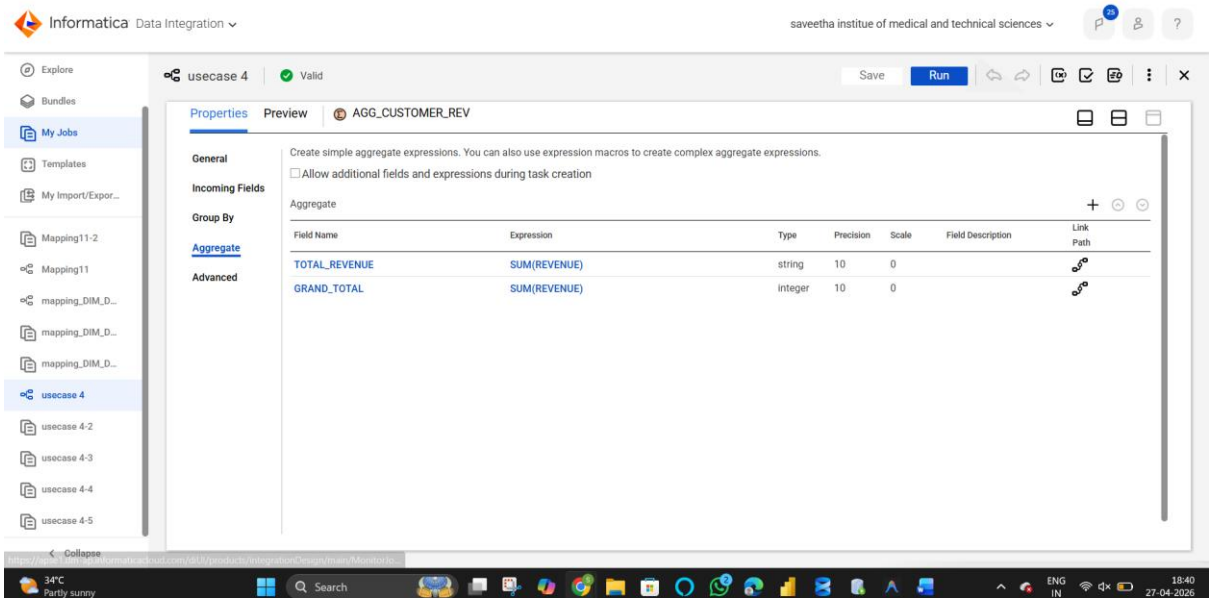
- To measure call activity and revenue
- To analyze performance by call type
- To generate business insights



KPIs Calculated

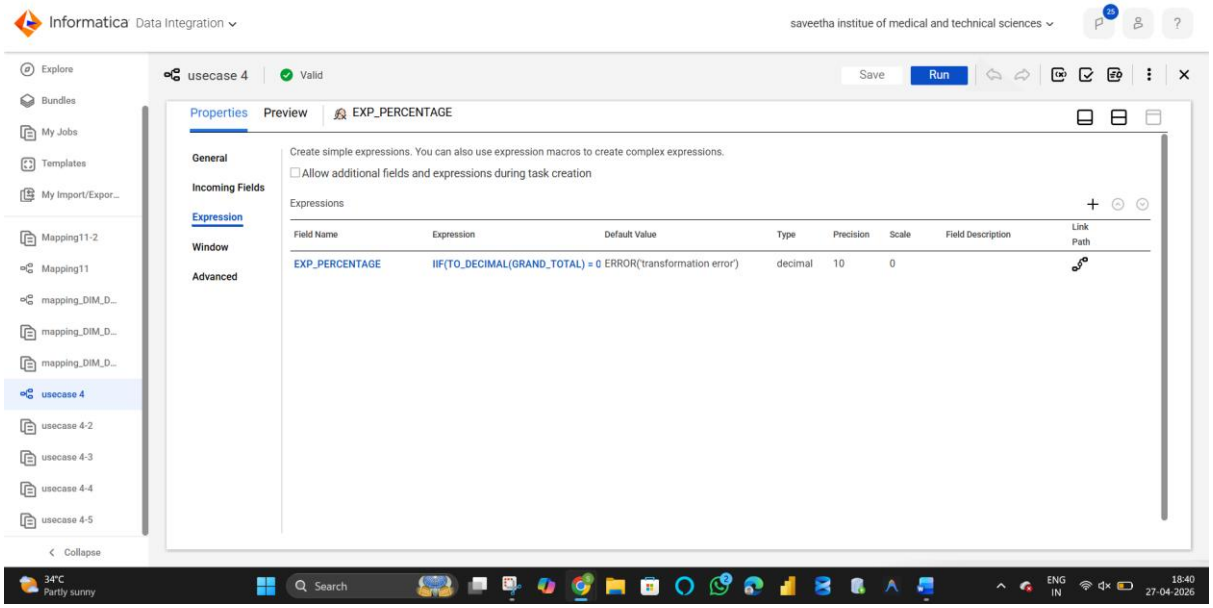
- **Total Revenue** → Sum of all call revenues
- **Total Call Duration** → Sum of call durations
- **Total Number of Calls** → Count of CALL_ID
- **Average Call Duration**
- **Revenue by Call Type**





KPI Calculation Logic

- Data is taken from fact table
- Aggregation is performed using:
  - SUM(REVENUE)
  - SUM(DURATION)
  - COUNT(CALL_ID)
- Data is grouped by:
  - CALL_TYPE_KEY
  - DATE_KEY



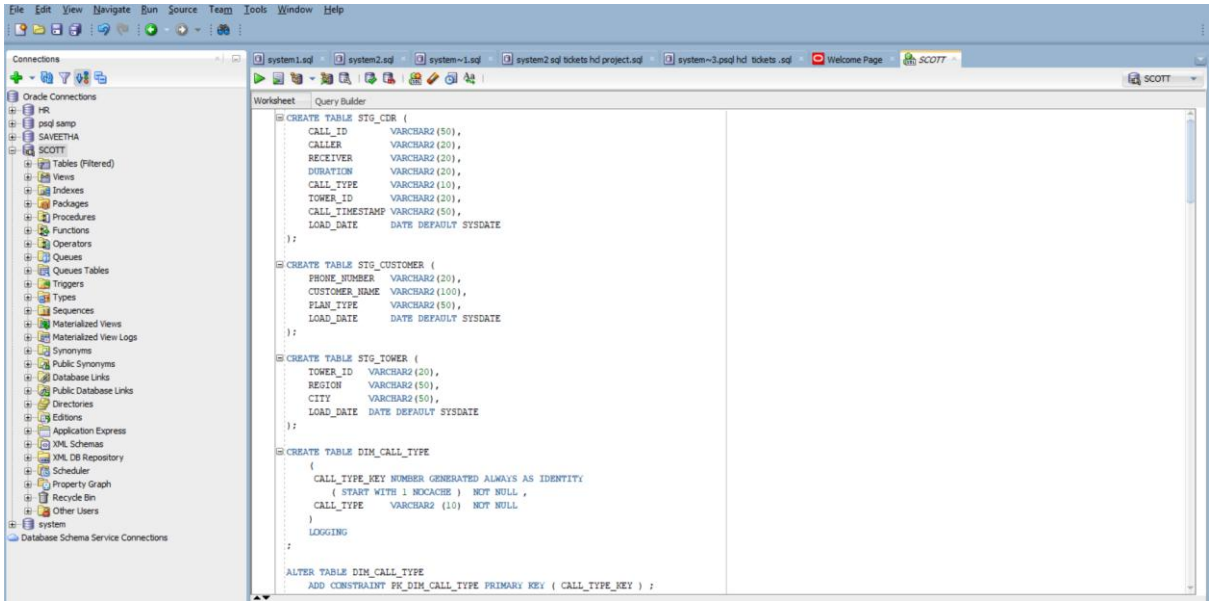
Error Handling

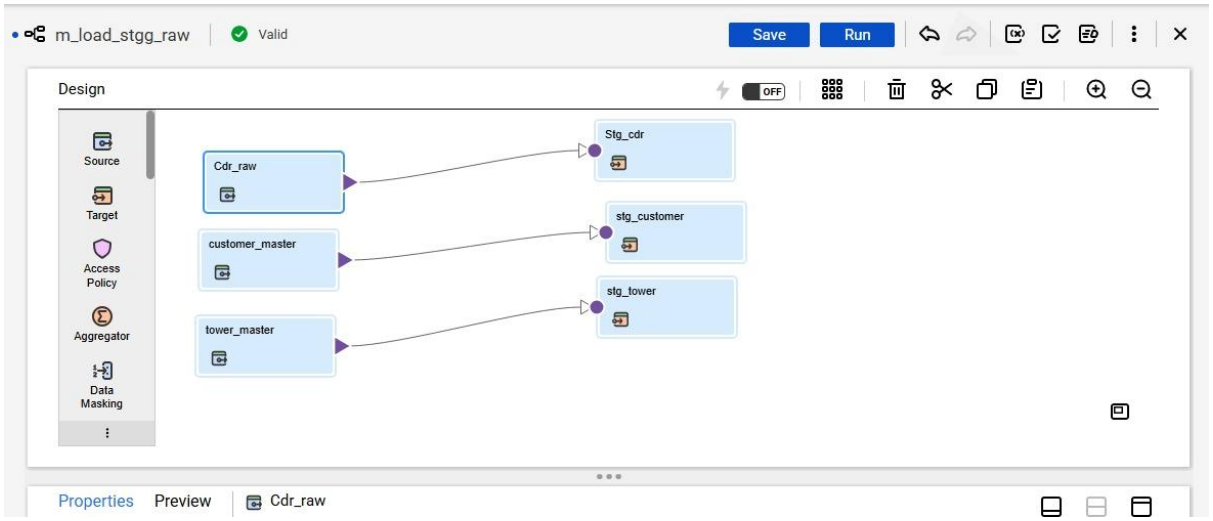
- NULL values are handled using default values
- Invalid records are filtered using Router transformation
- Duplicate call types are avoided using Lookup logic

10. Outcome

- Clean and structured data is stored in DIM_CALL_TYPE
- KPIs provide insights such as:
    - Which call type generates more revenue
    - Usage patterns of customers
- Helps in business decision-making and reporting

THESE ARE THE SQL QUERIES USED FOR CREATING TABLES

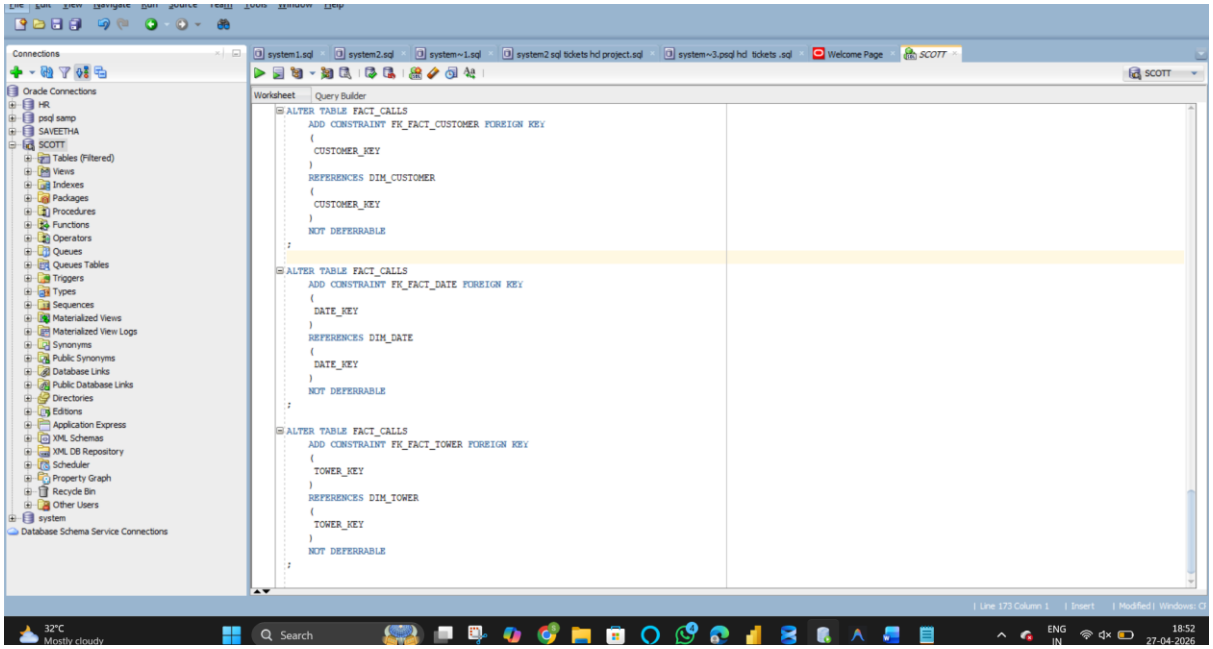




IN INFORMATICA STAGGING

```oracle
CREATE TABLE DIM_DATE
    (
     DATE_KEY  NUMBER  NOT NULL ,
     FULL_DATE DATE  NOT NULL ,
     DAY       NUMBER ,
     MONTH     NUMBER ,
     YEAR      NUMBER
    )
    LOGGING
;

ALTER TABLE DIM_DATE
    ADD CONSTRAINT PK_DIM_DATE PRIMARY KEY ( DATE_KEY ) ;

CREATE TABLE DIM_TOWER
    (
     TOWER_KEY NUMBER GENERATED ALWAYS AS IDENTITY
        ( START WITH 1 NOCACHE )  NOT NULL ,
     TOWER_ID  VARCHAR2 (20)  NOT NULL ,
     REGION    VARCHAR2 (50) ,
     CITY      VARCHAR2 (50)
    )
    LOGGING
;

ALTER TABLE DIM_TOWER
    ADD CONSTRAINT PK_DIM_TOWER PRIMARY KEY ( TOWER_KEY ) ;

CREATE TABLE ERROR_LOG
    (
     ERROR_ID      NUMBER GENERATED ALWAYS AS IDENTITY
        ( START WITH 1 NOCACHE )  NOT NULL ,
     CALL_ID       VARCHAR2 (50) ,
     ERROR_MESSAGE VARCHAR2 (200) ,
     CREATED_DATE  DATE DEFAULT SYSDATE
    )
```

```oracle
     DATE_KEY        NUMBER ,
     CALL_TYPE_KEY   NUMBER ,
     DURATION        NUMBER ,
     REVENUE         NUMBER ,
     IS_INTERNATIONAL NUMBER
    )
    LOGGING
;
CREATE INDEX IDX_FACT_CUSTOMER ON FACT_CALLS
    (
     CUSTOMER_KEY ASC
    )
;
CREATE INDEX IDX_FACT_TOWER ON FACT_CALLS
    (
     TOWER_KEY ASC
    )
;
CREATE INDEX IDX_FACT_DATE ON FACT_CALLS
    (
     DATE_KEY ASC
    )
;
CREATE INDEX IDX_FACT_CALLTYPE ON FACT_CALLS
    (
     CALL_TYPE_KEY ASC
    )
;

ALTER TABLE FACT_CALLS
    ADD CONSTRAINT PK_FACT_CALLS PRIMARY KEY ( CALL_ID ) ;

CREATE TABLE HIGH_USAGE_ALERT
    (
     ALERT_ID      NUMBER GENERATED ALWAYS AS IDENTITY
        ( START WITH 1 NOCACHE )  NOT NULL ,
```



11. Conclusion

This taskflow ensures proper handling of call type data through staging and dimension loading. It also enables KPI-based analysis for telecom operations, improving overall data quality and business insi